ActInf Textbook Group ~ Cohort 7 ~ Session 21 (Chapter 8) ~ 3/27/2025

TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 21 (Chapter 8) ~ 3/27/2025 | 1:02:31

all right welcome everyone we're in
cohort 7 the second discussion on
chapter8 so there's a few ways we can go
we can look at chapter eight in the
context of the book um in terms of
continuous time
modeling uh following up on chapter 7
from discreet time modeling also uh
after or just going to it I can share
some updates on continuous time modeling
from this morning in the RX and
fur group
where we um developed on the new example
from the RX infer team with the lorenza
tractor and an example with the neural
network so we could talk about that and
uh modify it in the context of
continuous time modeling
um first though let us just go to the
chapter does anyone want to just share a
section of chapter 8 or any question or
quote or or image or anything like
that I have a
question okay Andrew then
John uh yeah so they talk about um this
attractor like this point that things
tend to but they seem to talk about it
on two different levels like one in
reality and then one in the maybe the
generative model I guess like and so if
you could clarify that distinction um
how those work and what's their relation
yeah so when you specify the entire
simulation you will have let's just say
we're dealing with with some with

temperature in the room we could have
situation where just to to get into this
Y is our
observable X is going to be our latent
State x dot is the rate the change
landscape on the Laten state so this is
like the thermometer measurements coming
in on Y and the Laten State on X and the
change in the temperature um you could
have a room with a real attractor
setting like there's there's actually a
variable uh v such that when the true
latent state is exactly V the the um
rate of change is zero and when the root
temperature is higher it drops back to
the point of tractor and vice versa for
being
lower then you have the agents model of
the external
process that could be built to be
structurally identical like they're both
dealing with a point detractor in which
point uh the agent model will tend to
converge pretty simply to the correct
parameters but you could also have a
situation where like the real room is on
a limit cycle so it's just oscillating
or it's by stable but the agent's belief
is that there's a point
attractor so to answer it this is just
one example of attractors there's Point
attractors limit
attractors uh and so on and in practice
you end up specifying the actual
generative process that gives rise to
the agent's observations and then you
specify the agent's model of that
process and those can be structurally

identical or
not so so the attractor is basically a
construct of the a of the
organism in the way that you're
describing it like it's something it
would wish for you know which may not
exactly even exist there but may be
useful
nonetheless is
that yeah I mean on on multiple levels
there's there's the the modeler making
the temperature model of the room and
possibly just choosing to use an
attractor style dynamical model just
just as a map and then there's the agent
within the simulation using her istics
or constrained families to do
variational inference
in and and that could be simplified on
up to any
point and but the crucial thing is that
is the attractor in the agent's mind
that that there's something there they
ch chasing like they're trying to tune
into and at least of understanding and
for me that's that's very uh relevant uh
because I'm trying to do this like two
mind uh dialogue where like you'll have
a a continuous mind that's having some
kind of affordance or ability and then
the attractor would be like a concept
and then that could feed into this like
discreet mind let's say uh that you know
has connections between Concepts let's
say but so it seems like an attractor
can serve as what I would call a concept
and the process of approaching that
would be like some kind of skill or

ability or
affordance well yeah I mean in figure 86
this is like a classic hierarchical
architecture where the lower level is a
continuous time continuous stat space
model and the higher level is discreet
time discret State space model so it
could be like you have some sort of
continuous time Dynamic whether it's
attractor or just some sort of
oscillator and then it's like there's a
discretization let's just say into
quadrants so then even though there's
this continuous two-dimensional State
space at a lower level it gets
discretized into just a four state
discrete State space model that's
basically digitization so this
corresponds to a variety of settings
it's really convenient for taking analog
sensors and then discretizing the
information it's been applied to folk
Psych ology in terms of like embodied in
interoceptive processes which are
continuous in their nature leading to
discretization and action
selection okay good so that it has been
done that's um
okay cool um John and then yeah we can
just go to any any part on eight or we
can when we get to Len we'll look at
kind of Len
2025 well well this is what I wanted to
talk about actually because what I
learned about the lorenza tractor or the
luren system I should say I I I learned
that it was a very simple system that
isn't even very good for modeling the

the weather uh and so I'm wondering how
you could use this system in like all
sorts of other uh other circumstances
when it's not even that good at modeling
the weather which is what it was made
for yeah great great point so I'm not
sure if it was actually meant to be like
the best possible weather forecast model
meteorologist but the point was the simp
it was just it was just meant to be a
simple model that captured certain
aspects of the system right and kind of
the Breakthrough was that this
deterministic dynamical model in
continuous time exhibited chaotic
Behavior so that showed that really
simple models can't have complex
chaotic I don't think that was a break
through because um for example po K
looked at systems like that around the
turn of the 20th century
uh the I think I think this though might
have been the first use of the idea of
an
attractor but my question is if it's if
it's not useful for what it was built
for how can it be used to model even
other things like
um yeah yeah agreed Loren definitely is
building on different attractor Concepts
from po so let me just share a little
bit from RX and fur
um this were some some updates that that
I pushed this morning but based upon um
examples that they brought in just a
couple hours ago so six hours ago um so
again it it's not that the the Loren uh
model has just a few

parameters just here so if you wanted
performance you might not necessarily
use this exact model
but conceptu I've have a weird problem
with zoom I I I like I only see like my
uh my window like in the main screen I
can't see what you're sharing you
someone help me with this maybe like
right click
on uh can you see the shared window at
all right I I see say thing it says
Daniel fredman screen but I click on it
and it's very small and I can't read it
you don't see it at all or no I I do see
it but it's very small and I can't read
it maybe do pin right click and pin
it I'm try I'm right clicking but it's
not doing anything
um I'll try to figure this out yeah yeah
yeah it's all good but just it is there
so it has these several parameters and
it's a it's a sort of proof of concept
showing how with several parameters you
can get this dynamic chaotic Behavior so
what this RX Ander example did today so
it's in RX Ander examples. JL in the
advanced examples integrating neural
networks with flux JL so they uploaded
this six hours ago as a notebook which
is really useful for like web-based
viewing um and then I modified it into a
script and made some
modifications um so what what this
script does we'll just look at it in the
context of this notebook and then we can
go over to cursor and and um play around
with it um so first they Define the
Loren's system this is going to be the

underlying generative process so in in
uh terms of Chapter eight so kind of go
back and forth between chapter eight and
this more recent work the underlying
process is the
X why are going to be the observations
that are sampled from the Loren system a
data set is created with noisy samples
coming from this Loren system so here's
this image this is kind of like the
gerer boach book cover we have a
three-dimensional continuous time
dynamical system with Loren's Dynamic
and then here are three of the
projections onto each of the par wise
axes so there's like we have some you
know some insect flying around in a
threedimensional room and these are the
projections with a shadow on three of
the walls
um
so we we are using this because the the
noisiness and the underlying chaotic
nature of Lauren system make it hard to
find a convergent parametric estimate of
the transition Dynamics so it's it's
emblematic of complex adaptive systems
like weather or biological systems that
might dwell in a reg
and then have some kind of switching
into another regime again showing how
that comes out from just a couple of um
of equations which are
deterministic but the sampling from them
is not any chaotic system is going to be
like that right so so why
Loren yeah because it's classic and
there are other ones this is one with

very very few parameters so like there's
the double pendulum there's a variety of
chaotics systems to study like
empirically and analytically and this
more of a
metaphor no we're actually studying this
model what do you mean
well um if the if you have a chaotic
system and you're representing it with
the lored system like that that wouldn't
be good unless it actually is modeled by
the lored system right well you can have
a toy model right like you can study a
toy model just for lots of reasons
yeah there's the pure pedagogical
element which is like if you don't have
this Lego block in your
grammar you won't have a key piece of
understanding how it works in this case
and then if there's a different model
that fits the data better when you're in
like a data fitting situation like
chapter nine right you might use a
different model or so is the reason you
use LEDs because it data fits is that
the is that the answer no because it's
like has a long history it's simple and
it gives us an intuition on these
chaotic switching Dynamics with a lot of
um like Rich behaviors from a simple
underlying just a couple equations which
we can Define in a few lines of code
okay but so then there's like a world of
active inference that works with chaotic
models besides Loren and Loren is just
in the textbook is that is that the
thing yeah you like if you put a
different model here the rest of the

notebook you could just this is just an
example so if you have a different model
or you want to change how you can put a
different model here just just wait to
understand what's what's happening in
this notebook so a simple model just
keeps things short and easier to like
Daniel said if it's classic then people
understand there it's easier to
communicate yeah there's so and it is
one of the simplest I mean it's just
these
three dimensions and three updates okay
so here's noisy samples coming from
Lorena tractor now we want to
infer the lat and state given the obser
the
observations so flipping to chapter
eight that we are getting observables
and we want to infer neural States or
latent States like in SPM statistical
parametric mapping so kind of the
precedent to FP and active inference
you'd have like neuroimaging data as
your observables and you have underlying
neural activity as Leon States
okay however the hidden dynamics of the
ren system exhibit nonlinearities and
cannot be solved in the closed form one
manner of solving the problem is
introducing a neural network to
approximate the transition Matrix of the
Loren
system so note that the notation is
different in this example than it is
from the textbook so in this example a
is going to be the transition operator
and B is going to be like the ambiguity

operator that Maps between the latent
States and the observations whereas in
the textbook it's like the other way
around B is the transition states in the
latent space and a is the observation
Mountain
okay flux. JL is used as a neural
network package in
Julia here's the RX andur probabilistic
model we could return more to this or or
people can learn more about RX and fur
or get involved in the project but this
is a very simple State space model with
just five
variables um
first the probabilistic model is used to
extract parameters from the untrained
neural
network and basically it does
terribly like the untrained neural
network doesn't predict the data at all
um so it was just randomly initialized
and so the probabilistic state space
model also just this green line it it
has nothing to do with the
data then the neural network is trained
using a free energy objective which is
how neural networks are
trained um here's just neural network
training
code
then we get here which is showing that
we can use the probabilistic model the
state space model like the explicitly
defined RX and fur model to recapit
ulate the parameters of the neural
network and so I took that several steps
further um it's super super exciting

because this is showing a a really
critical design pattern which is about
fusing the complimentary strengths and
functionalities of the empirical data
fitting highly parameterized nature of
like machine learning and neural
networks
with explicitly defined and constrained
probabilistic models like in RX and
fur so basically they have complimentary
strengths neural networks are Universal
function approximators they can
interpolate any given function you know
given enough training and size and all
of that um however they have high
computational resources they have
challenges with interpretability and so
on they don't have genuine uncertainty
estimators all of these different kinds
of
limitations here's showing a a way in
which the parameters of the lorens can
be inferred with a neural
network and trained on the data just
like anything else in machine learning
or just sort of computer science today
and then we can infer
transition
parameters within a very very tightly
defined probabilistic model which is
small enough to be defined within this
app model Block in RX and
fur so this is kind of it's It's you
know things have happened since 2022
this is one of them from just a couple
hours ago um and it's just really
exciting because it shows like you can
do pure endtoend

probabilistic programming on base graphs
to directly fit parameters of
distributions from data and and that may
be adequate that might might be direct
and awesome but this is a super
interesting pattern that can be added
into a lot of extent machine learning
pipelines which is or language model
pipelines where you have some kind of
reinforcement learning or neural network
system and then you use explicit
probabilistic modeling on top of that to
gain interpretability
genuine uncertainty estimators and other
kinds of benefits like lower compute
resources for these probabilistic
models
John uh so uh are you or is anyone
familiar with step wolfram's concept of
computational
irreducibility yeah so um you you said
that the the AI model can match any
function how does does that work with
computational irreducibility or uh are
we rejecting computational
irreducibility what's how does match I
said interpolator or approximator so
yeah like you could have let's just
whatever the actual ontological
computational irreducibility is of like
the weather you can train a linear model
to approximate it it might be a terrible
approximator your startup might fail you
might pick the wrong weather you know
Etc ET Etc but you can approximate with
a model
so those Blobs of irreducibility in
nature can be

approximated using statistical models so
like to take this into sort of like the
halting problem like you can't know
whether a script is a computer script is
going to like what its output is going
to be for some kinds of scripts but you
could have a statistical estimator that
says I I think there's an 80% chance
this script is going to Halt without
running
it or you know you human behavior could
have these elements of computation or
irreducibility but that doesn't mean you
couldn't make a statistical model as a
behavioral
scientist so it's kind of like map
territory like irreducibility is about
the territory but you can always make
statistical approximators but it does it
create a limit on how good the
approximation can be right like if the
approximator becomes
perfect then functionally there was
computational reducibility because you
could reduce the underlying process to
something
simpler but to the extent that the
approximation is not perfect then that
computation has not been reduced now
whether it's in principle irreducible or
just the modeler hasn't tried hard
enough is a second
question but like here's here's an
example of the advantages that you get
from building this probabilistic model
so here is like on a given Dimension so
just for the x value or something of the
luren attractor we get a a moving

posterior distribution with a mean and a
variance but the neural network won't
give us a mean and a variance it's just
giving us its best prediction of the
location at the next step whereas this
gives us like a moving basy incredible
interval so the these these kinds of
methods are are complimentary
um and it I mean just again within the
last hours this has greatly expanded our
toolkit because it gives us a access to
building probabilistic models on top of
trained machine learning
models rather than needing to do
endtoend training only through the
probabilistic
model so could you maybe expand on that
Point like where you said with the mean
and the variance that uh I guess you're
getting this uh data coming in from a
neural network and then you're able to
you said I think statistically
approximate that from the other end like
from top down
or yeah yeah so like let's just say that
that we have a language model and we're
going to ask it um how many people live
in x Place MH and it is going to do this
forward pass through billions of
parameters and then it's going to say
like 11 million 10.5 million 13 million
17 million like it just it's outputting
these numbers which represents in that
run with that temperature the likeliest
outcome that's how it works with
sampling the likeliest outcome given
some some noise that's added back in we
could take that sequence of

observations and then fit a model that
just fits a mean and a variance from the
language model's
outputs so we kind of infer like the
true value for its belief about the
population of that
region from sampling its
outputs so it still could be wrong
relative to the real reality like let's
just say all of its training data
suggested that there's this amount but
then more people are there so this the
this isn't to say that it's veridical in
some future moments but inferring the
latent State and having an uncertainty
estimator on the lat and state is always
going to be more accurate than any given
single measurement it's like the
measurement of you know if we have the
mean and the variance of the height of
the children in the classroom that's
always a
better um number to work off of than
just one measurements and and I think
that's a nice example of what this whole
setup is that the statistics kind of
makes this presumption that there's
something meaningful that we call the
mean and the and the variance but the
reality might be that this this neural
network is just all over the place and
that's just there's no really I mean it
that means nothing but from in from a
statistical point of view in terms of
corralling or shepherding or like just
trying to you know get get your H our
hams around this it's very useful and it
is practically use ful even though it

from a meaning point of view it might
not actually mean anything but it
suggests what a meaningful meaning would
be like that if it was if this is a
meaning you're supposing hey like is
this what you're talking about like is
this what you mean the machine would say
I don't know what you're talking about
but that's how I'm thinking about it
yeah so MB you mentioned balance let's
just see if cursor will uh will bless us
today okay what's kind of a
question that we could ask about
balance and see if we could be dealing
with almost this two minds Paradigm of
like a neural network that's fitting
noisy proprioceptive data from a
continuous dat space and then like a
conscious explicit mean and
variance estimation from there but like
MB was an example of a setting like that
or what would be a question that that
relates to the vestibular Coke
CER yeah um C could you help me
understand that a bit better as I I'm
I'm yeah trying to follow uh and and
yeah you can just call me Mark um okay
okay yeah mb's cool for Markov blanket
too oh yeah sorry I didn't see that my
uh my screen name is that Markov
blanket um Yeah so basically like using
um cursor as sort of this this um
augmented coding setup just it it may
not work within you know the the 30
minutes we have here but just just to
see possibly um that's what these
examples are great for is like starting
with this

structure and then modifying tweaking
the example like the neural network
could be flipped out for a more
sophisticated neural network or here in
this situation
let's see if we could adapt it to study
some question of Interest with like a
vestibular clear balance
question yeah so are we sorry
um specifically so we're trying to put
data into this model is that right we're
trying to build a
model using this this program is that
correct yeah yeah modify this the the
architecture of this model where we get
noisy data from a continuous dat space
you know chapter 8 and then we're going
to infer the we're going to use a neural
network to train on the parameters from
this noisy State space that could be
liken to our
subconscious um you know
embodied predictive networks and then
infer
parameters from a probabilistic model
put on top of that neural
network okay so if we're looking at some
kind of uh data like um and tell me if
I'm if I'm understanding this correctly
like postural sway is that would that be
one of
the data that we're looking
okay okay the noisy data are coming from
proprioceptive continuous State
spaces related to postural sway
and and that and that could be like
modeled as Loren attractor with with um
chaotic transitions between

qualitatively different
postures like right the threedimensional
stat space could be um the three angles
of your
finger so then there's like a Zone
within that state bace that the
finger the finger's biomechanic ICS are
related so it's like and this again to
kind of get to this like what's what's
the distinction with active inference
and these other neural network or other
kinds of models what would be the
approach to making biomechanically
realistic finger
movements well one approach would be we
get hundreds of hours of movement of
fingers and try to do endtoend learning
based upon recapitulating realistic
looking videos of hands for example um
but obviously even with billions of
hours of
video making images with realistic hands
is still challenging whereas this
complimentary approach would be you
actually have this explicit
parametrically defined model of the
biomechanics of the hand and then you
understand well here's the zones of
State space within that defined model
that the hand can be in and then will
generate images of the hand from these
biomechanically realistic
postures versus just trying to
recapitulate the the
appearance from real
video and so
so okay and then what what what kinds of
question would we want to address with

that kind of
model so are you I mean I could say
something like you know one's uh
conscious perception of threat you know
some something like that would would
that
be is it fair to so this is all new to
me right but but taking someone's uh you
know conscious
perceptions um
and looking at the effect on on the
other end of the you know
entropic dimension of their postural
sway characteristics you know some
something like that that that would
be I I don't know if that's the the
right kind of question
to to ask for for this type of exercise
but you know that would be an example of
something that would be of interest I'd
like to add a related example uh learned
about uh when I went with my friend to
his son's aido practice yesterday and
there was a um he told me about ban Jean
so there's this martial arts it's all
about these different hand movements I
there like eight different ones so like
they just cycle through them and so you
can um so it's very much like these
attractors like you can think okay my
hand's going to be grasping or my hands
going to be blocking let's say or my
hands going to be striking and so in
each case I kind of picture I think they
even kind of suggest that that there
like if your hands grasping you have a
particular attractor that you're
projecting and so as you move your hand

you this this attractor is guiding the
grasping so that in that context you
would you would be imagining this
grasping and and you would be centered
on that and it would all work out but
then in this Kung in this B ban School
based on the iching eight trigrams then
you might go to let's say striking you
see and then you might go to let's say
blocking so the context keeps changing
for what you what is an attractor and
you just C cycling through all these
attractors it's it was like bot
periodicity I talk to Daniel sometimes
about but I thought that was kind of
curious that wonder if like that could
be connected here yeah like um this was
uh it it relates to chapter chapter
eight like in this page on uh this is we
we did a book stream with Thomas par a
couple years ago and uh this part right
here relates to basically recurrent
attractors and continuous State space
model so initially the active inference
models were more like generalized
filtering then people wanted these
sequential Dynamics so that's like dealt
with in chapter eight like cursive so
then it was like there was models where
it was like the limit attractor was like
a cursive letter s so it's just kind of
like drawing a cursive letter S and the
continuous Dynamics were basically like
the arms joint mechanics on the table um
but
planning on digital computers is much
easier with discrete State space models
and so that is what led to these hybrid

architectures
where there are mixed continuous and
discrete State
spaces because planning is is more
easily considered in the discretized
setting like a chess planner
um so it's like with the eight hand
postures um it's like that there's eight
discrete options at this higher level
and then action selection you know
context switching attention which is
modeled as internal or covert action
meaning it's not observable from the
outside attentional regime shifts within
this discrete State
space um is you a descending motor
contact switch such that the body
engages in a different limit attractor
so it's like raise your leg now put your
leg down now raise your leg now put your
leg down so it's like there's a
biomechanical
continuous
attractor in that case it would be
basically like that would be the
example from here
move move your point okay point at the
top corner of the room not point at the
bottom corner of the room not Point At
You Know point over there then there's
this transient
Dynamic new set point is established and
then um error error um reduction with
propri receptive feedback no touch your
right wrist and then well once you do
that that the proceptive error is
zero when you have when you're when
you're doing that the top down

attentional regime I should be expecting
Pro receptive feedback reflecting that
I'm touching my wrist and the ascending
propri receptive data have no error term
and that's a fixed point or the fixed
Point could be oscillatory like wave
your arm like this so that's like an
oscillatory
ATT
tractor one thing I've I've come to
understand is that as we so so use the
example of moving your arm to a a
point in order
to initiate actually to plan and conduct
that movement there has to be sensory
I'm assuming that's propri receptive
attenuation is is that
right and so how do you then use the
appropriate receptive input to
understand the movement in the model
when I think what we do is and you can
absolutely correct me is we turn the
Precision
down as we attenuate the sensory in to
in order to to overcome the the evidence
right that that I'm not moving to use
that example does that make sense yeah
sensory continuation comes up in five
chapter 5 as well but yeah exactly like
modulating
attention so what are the two ways to to
reduce free energy there's change your
mind and change the world you know
updating beliefs that's changing the
Mind through learning an inference or
updating action to to to modify the
world okay and what is going to
control whether the agent engages

in passively updating its parameters now
parameter updating is its own kind of
action but not gross action you know or
changing through through action and that
relates to the attention that's paid to
the ascending predictions
basically so that is is empirically
observed in postural settings as well as
like Cades which is like kind of like
micro postural I mean it's it's ocular
postural um that
basically in order to enable a
move if you kept High Precision like for
example all my perceptive is that I'm
seated in a chair so how is it that I
can get out of a chair when it's like no
I and and in reinforcement or reward
learning actions are selected based upon
their expected reward whereas in active
inference we're doing direct policy
inference on the likelihood of action so
it's not that standing up would be more
rewarding for me it's I expect myself to
be standing up so it's the Fulfillment
of likelihood rather than the
maximization of reward but how could it
be when it's an obvious
counterfactual how could it be likely
for me to be standing up how could I be
the kind of guy who's standing up when
my vertical appropri reception is that
I'm not and the answer is transient sens
attenuation makes it suppressing the
proprioceptive
input enabling there to be more
ambiguity of which position I'm actually
in which creates a kind of like postural
liminality which then a top- down belief

no actually I am standing up or I'm on
the path to standing up can sort of get
the ball rolling over a critical
threshold that then action fulfillment
can take all the way that's been modeled
in like situations with like hyper or
hypo
dopaminergic cases where like somebody
may have challenges initiating movement
but once movement is initiated it
continues or they can initiate but not
complete a
movement so it's just that transient
suppression I think that was that's the
key is that it to initiate and then
you're on to
some level processing that propri
receptive feedback to to help with the
movement itself right when the movements
in process there can't be this
continuous suppression it has to be just
I picture it just kind of as a as a
Flicker and then you're gradient a
gradient of suppression like yeah well
said you become more and more prec
demanding and more certain and more
precise yeah or or it's like driving on
a windy road and it's like on a straight
part you want to accelerate but then you
you relax off in order to update your
vector so it's like when your gaze is
fixed you want to be having high
attention to the visual input but then
during IM volitional or sub volitional
eye
movement visual attenuation occurs which
is why we don't get motor you know don't
get nauseous outside of like vertigo

type settings with eye movements because
when we look away like we get this
attentional blink we're actually not
accepting or paying attention to the
visual input even though photons hit the
retina the same rate during the movement
of the eyes so that's like kind of like
it's like depressing the clutch on
attention shifting into a different spot
and then we re-engage the Precision and
now we're like getting information from
over
here i' say there's good evidence for
those who for folks who go on to um have
you know something like persistent
dizziness is that there is not uh good
uh sensory attenuation with movement in
fact there's there's this excessive
Precision that seems to be a theme among
you know a variety of uh
conditions yeah that like that is sort
of like
that is what Precision Psychiatry and
all these kinds of approaches get
towards which is like sometimes it's
that there's a um it can be an
uncertainty estimator that's too precise
or not precise enough that leads to
different behaviors so it's not a
question of just like moving a variable
up or
down if I was to refine that question
that that you kindly asked me to to try
to formulate it say something like um
were the effects of excessive tension to
postural sway you know which would be
the same thing that would probably occur
during a a you know a perception of

threat um but what are the effects on
the actual sway parameters so this just
becomes this ongoing Loop and that would
be you know something like to to use the
you know the tibular phraseology like
Mal debarkment which is just like an
excessive um duration of something like
SE legs
yeah I don't know if that's something we
could put into a model like
this yeah like the it's that's kind of
that that is the exploratory modeling
and inquiry is like okay so we have this
model that's getting trained on Pro
receptive data so you know in progress
but then we could say something like
well but let's just say somebody is
acting based upon their parametric
estimators of posture
what would happen if their if the
Precision was low so let's just red
let's have it let's have the the best
fitting model and then let's simulate
movement from someone whose Precision is
three times higher or three times
lower and then that you
know yeah Andrew did you have a yeah I
wanted to jump in with an idea that may
actually be related to this but that
I've been having in this situation where
let's say you have this task that you're
given a triangle and you're going to put
in uh and you're kind of blind but
you're going to put in a little circle
and you're try to expand the circle
until its maximum size and you can never
be perfect at that but you you try what
you can so you're given some triangle so

in the beginning when you do a task like
that you're going to be bumping into
sides or whatever like you don't know
how to guess what would be the right
Maxim you know the right amount to
increase the radius Etc but at a certain
point you'll get the answer or something
rather satisfactory and then I imagine
there's like a replay in let's say in
your mind to say look okay this is what
it took for me to do this task but now
that I know what the answer is where the
center is I'm interested to learn from
that by saying okay now that I know
where I end up I would replay it in my
mind to do that as efficiently as
possible like I'd make the right guesses
all along the way
and so if you do the Replay for the
cases you've done
before um you get much more efficient
algorithms that you're developing so
let's say you're getting you're given a
whole bunch of different triangles and
so they're different conditions and
whatever but I think this notion of and
I think it goes like to the statistical
um interfacing that you were doing
Daniel saying okay it's not a particular
answer but like you're going to get a
statistical mean and a variance for like
what's optimal Behavior you know what's
optimal presumption like where is this
center of the triangle probably most
likely going to be if if I'm if I'm
thrown into some kind of triangle yeah
yeah like live this notion of Replay
that you you would want to replay and

get your optimal strategy and then you'd
want to go into a new problem with that
optimal strategy and keep improving it
yeah so like live stream 51 series with
isura at all this is a really critical
paper that shows that there's a
monotonic relationship between the loss
function used in neural network fitting
and the variational gradient descent on
the semantics of the base graph across
the blanket and they use this
postdictive learning
strategy where basically at each time
point you you minimize so the factor
graph with fictive causality of factors
MH so decision making to minimize the
future risk
so we want to minimize future risk for
the unocc herred event and the way
that's going to happen is by remembering
our past sequence of actions and
observations and then
reassessing what would have been the
least
regretful updated risk
assessments given all that we've seen
till now or given within the sliding
window of what we've seen till now so
then it's fictive causality is
propagating backwards in
time in the sense of the the agent is
saying if I would have done that given
what I know about this now this is how I
think it would have occurred and using
those re-entries to reduce upcoming risk
and it can
be moment like very rapidly after the
final solution so like with posture I

can just imagine like let's say you feel
wobbly and you kind of like regain your
balance but you just say wait wait wait
like I had it all wrong like this is
what I should have been having in my
mind so because so and if you don't have
that you're just going to be wobbling
all the time and you're going to fall
down but if you're wobbling a little bit
and you're able to reparameterize well
what is the my proo SPC
situation then you'll be you'll feel
stable that was just a momentary
thing that was somebody said said
something about about
what I was just saying you could you
could also do that on on a false
inference right like um like an
overattribution of something like let's
say you you have a really and we see
this in the clinic you have really kind
of fluky circumstances around a fall you
know that you couldn't possibly you know
recreate sort of a oneoff experience
that somebody has and then they go
through the process of
it sounds like mitigating the likelihood
that that will happen again and they do
that to a a level that
is excessive right so then their
subsequent movements become very
constrained like shuffling their feet or
something yeah as for example or they
have a very like tight I guess you'd say
um you know Excursion of of their of
their postural sway you know so so those
kinds of things and and they're same
time there's a a greater conscious and

this is what I'm interested in is like
you know the the the conscious over
tuning of
[Music]
precision yeah like the I mean that is
is definitely where nested models would
come into play like an injury
occurs and then there's increased
attention to to the pain or there's
modification and that sets up a new
attention with descending and propriate
exception
ascending and those can be those can
result in a new
attractor W with reduced Mobility or
whatever it may be and that can
generalize to to other postures and and
even probably
other
circumstances yeah like again that's
sort of the toolkit and that's where the
textbook gets us it's just like hey with
with single layer models
this is the there's already these
richness of cognitive patterns to
explore and that's what chapter 7even
and chapter eight are about so you know
chapter six is the recipe these are the
questions to consider like going through
these is is qualitative initially and
then chapter seven goes into the discret
time models chapter eight goes into
continuous time models chapter 9 goes
into Data fitting but it's like that
that kind of is
one structure of the
process and then it's maybe you want the
single best performing model for your

deployment or or for your paper or or
there's also a lot of value in building
these portfolios of models and looking
how different features like
modify
uh you know explain different components
of the variant especially if it's going
to be something involving people where
there's like intersections of of complex
psychological
phenomena so it's it it it may not it
may not be just a single layer model but
this is some of the toolkit and knowing
what what a single layer model can
do
helps give give some of the um motifs
kind of like the rudiments for building
more sophisticated models and I just
it's an exciting day because the
applicability of these probabilistic
models to do parametric inference from
neural network models which are just
everywhere has increased markedly in the
last like eight hours which is pretty
fun to jump in like to think of these
two minds one continuous and one
discreet and I think of like the
continuous mind you know you want to
extract the critical point so it could
be like the Peaks or the troughs or it
could be like you know going up your
tensing your muscle you know and then
going down you're relaxing it and so you
have these different modes that then can
become discreet and I guess what I
learned today from you was that in the
other direction it's like projecting
statistical uh presumptions like oh

there's got to be a mean let's say
there's got to be a variance that's
attracting even if they don't actually
you know exist in any real uh physical
sense but they exist in a sense of
trying to get a grip on what's what you
you know getting a
handle yeah I mean and and and this this
like to the early questions like the
structure of the generative model
assumed by the agent does not have to be
the same as the environments the
environment could be like a bathtub
probability distribution like it's
either like dwelling low or dwelling
high but then the agent could be trying
to infer a gaussian so then they could
having all these pathologies where first
it's super tight low and then it's super
tight high and then it's in the middle
and it's super blurry and it's swapping
between them or it's staying in one so
it's like it's like they're fitting
Ultra well when it's on the low with
their High Precision low estimate but
then they're completely confused when it
flips to the high state so it's like
those are those are the kinds of
Diagnostics that you want to look at to
understand whether you're dealing with
the wrong family of variational
distributions and it's like you could be
walking uphill for a ways and then
walking downhill for the ways like your
posture is complet like there's no
average that make it's a it's a
non-concept to say there's some kind of
average there because depends on where

you're walking and you don't know where
you're going to be walking but but the
idea that you have some kind of
momentary balancing you know
parameters that you're kind of pre
guessing like that's they don't have to
be real but they could be meaningful I
think yeah and and it comes back again
to the uh Precision modulation and atten
ention mhm which is just over the the
papers and the years is is always
returned to as sort of a a Hallmark of
intelligence you know you want to be
precise when it's good to be precise and
to be open when it's good to be
open but when to do that is the question
and that's that's like a third mind type
of thing balancing those two minds I
think exactly agreed with
that um just in our last minutes let's
see how that
um what this what it will do for its
second output but anyone have any last
comments or
questions I have a question but it's not
about chapter
eight is it is it possible to change
your generative model like are there
there models that change models or yeah
yeah that's called structure learning is
when the topology of the model the
structure of the model is itself
modeled so yes then
so any other
uh I'm glad John Mark Daniel that to
hear from everybody was very helpful
today thank you
cool thank you for uh allowing me to

participate and and learn this is this
is great cool yeah I'm also just really
curious to see where where this last
training step will get us in the outputs
I mean just and I hope like again it's
not all about
code but starting with
examples and modifying them and kind of
staying on a ridge of
functionality is super powerful right
now like you can also this RX INF for
you're active in this
or um it's an open source project we
have a pro we meet weekly okay you are
in this m i mean it's lazy Dynamics and
uh reactive bays are the
developers but like there's a lot of
participants in The Institute who also
contribute and so and we're partnered
with
them but it just using cursor or similar
methods again so it isn't all about code
it's not all about reducing it to
code but I this shows a little bit and
there's there's other places to see it
as well you can get you with one prompt
you can get pretty far but modifying
what's already
functional is ridiculously
powerful so that's one thing that's oh
sorry sorry goad I was just saying it's
becoming a bit demystified as far as
that that was the one question I had
coming in is okay how do you take
something that's conceptual and then
take real numbers and and run it and it
sounds like it's it's it's kind of like
well you you just start with something

and then you say okay what if I increase
the Precision here or I change the model
there right that's kind of what we're
getting at is you just start to apply
numbers in a in a relative sense is that
right if if the and if even better if
there are real data but I don't know how
you would give um you know Consciousness
you know how you would give that numbers
you know like like attention you just
were playing with that zero to one as
far as the Precision of within the model
is that what you're doing yeah I think
like obviously Consciousness especially
if it's defined as unobservable is is a
perennial measurement issue in question
um doing attentional modeling and more
of these like um just cognitive
variables that aren't necessarily
related to self-awareness or or a
Consciousness those are more tractable
self-report is a
classic um so yeah Consciousness related
modeling is often tried and very um
fraught but but this
is I I don't mind it being FR so this
distinction between the continuous and
the discreet and then realizing like you
were saying uh Mark that sometimes you
shouldn't overthink it right like you
should be uh doing you know your
continuous movements sometimes you sh
it's just like your lucky mind when
everything is going well just keep
walking but you have the Unlucky mind
like hey there's a hole like don't fall
in right so so do you use your lucky
mind or your unlucky mind um uh that is

what Consciousness is all about like to
make sure they have the same information
and then deciding yeah but here I
shouldn't overthink it but oh no here I
should be careful so I think and see to
to the things we're learning like the
relation between those two minds one is
that like oh let go of the break in
terms of certainty and pretend you don't
know and then tighten tighten tighten
Titan so that shows you that process
shows you the direction of those two
minds that you're going from The
Continuous one that needs to be moving
to the discrete one that needs to kind
of like focus on some
attractor so those types of things are
indicators of um the Consciousness in
play I would
say let's see if it will output the
stick figure as requested but but you
know these functions are only a couple
uh so this could be like this is
pointing and like you're having your
hand waver and point at
something and then like for example
blood sugar is related to the Precision
of the ability to point if you're if if
somebody's having a a blood sugar crisis
they're going to have different
Precision of their
movements that's something that's more
empirically observable
able and then you could differentially
parameterize more overt
elements let's see if it it if it did
it absolutely
hilarious these are stick figures yeah

true posture propri receptive perceived
posture
so that gets super interesting right
like there's the real
body and then there's like these ghosts
of like what we predict and what
is and then you just say okay now modify
it so that it's taking in this data
instead of the simulated Loren
data so say which which what are the
three the true the true posture is the
red one the proprioceptive signals are
these noisier gray ones okay and then
the perceived posture is the blue one
that's that's exactly what people
describe is you know at some point okay
I realize my postural sway as normal and
that that almost takes some convincing
and
then but it feels like I'm you know I'm
doing this it feel and you've probably
had that experience you know um coming
off a boat or you even sometimes
transiently coming off an elevator you
you perceive that excess and for some
people there seems to be a a difficulty
attenuating that and and you know
returning and and to use your uh example
like you're you're walking up a hill and
then you're walking flat but like when
you made that transition you're still in
a sense like running the program as if
you're walking up the hill and now you
have this this
conflict
yeah that's the that's fascinating
looking at that that that's really
exciting a really great way to

uh to

Tom yeah it's open source take it and

and run slw walk with

it not too fast or stand and sway yeah

epic though it's epic okay thank you all

see you later bye thanks so much great

bye bye